

Health Economic Implications of Artificial Intelligence Implementation for Ophthalmology in Australia: A Systematic Review

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Purpose: The health care industry is an inherently resource-intensive sector. Emerging technologies such as artificial intelligence (AI) are at the forefront of advancements in health care. The health economic implications of this technology have not been clearly established and represent a substantial barrier to adoption both in Australia and globally. This review aims to determine the health economic impact of implementing AI to ophthalmology in Australia.

Methods: A systematic search of the databases PubMed/MEDLINE, EMBASE, and CENTRAL was conducted to March 2022, before data collection and risk of bias analysis in accordance with preferred reporting items for systematic reviews and meta-analyses 2020 guidelines (PROSPERO number CRD42022325511). Included were full-text primary research articles analyzing a population of patients who have or are being evaluated for an ophthalmological diagnosis, using a health economic assessment system to assess the cost-effectiveness of AI.

Results: Seven articles were identified for inclusion. Economic viability was defined as direct cost to the patient that is equal to or less than costs incurred with human clinician assessment. Despite the lack of Australia-specific data, foreign analyses overwhelmingly showed that AI is just as economically viable, if not more so, than traditional human screening programs while maintaining comparable clinical effectiveness. This evidence was largely in the setting of diabetic retinopathy screening.

Conclusions: Primary Australian research is needed to accurately analyze the health economic implications of implementing AI on a large scale. Further research is also required to analyze the economic feasibility of adoption of AI technology in other areas of ophthalmology, such as glaucoma and cataract screening.

Key Words: artificial intelligence, computer vision, finances, health economic, machine learning

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INTRODUCTION

Health care is a resource-intensive sector that requires a large portion of the gross domestic product of developed countries. In 2021, global health care spending reached \$8.8 trillion, which accounted for over 10% of global gross domestic product. This expenditure is an increase from previous figures of \$7.8 trillion in 2017 and \$8.3 trillion in 2020. Global health care expenditure per capita in 2019 came to \$1129.¹ Future predictions forecast continued yearly increases at a greater rate than overall global economy growth.² Health care expenditure in the 2019 to 2020 financial year totaled A\$81.8 billion, constituting 16.3% of the Australian Government's total expenditure for the same period.³ Additional costs specific to ophthalmic care, such as intravitreal injections, contact lenses, eye drops, and procedure costs, further compound the already burgeoning bottom line.⁴ The development of additional cost-effective strategies to deliver health care is an ongoing and increasing unmet societal need. Artificial intelligence (AI) may be able to facilitate this efficient delivery of care in ophthalmology.

AI is an umbrella term encompassing a group of technologies able to accomplish tasks without human assistance.⁵ AI is a dynamic emerging technology with many potential applications, including in health care. These medical applications include diagnostic, epidemiological, and screening use cases. As has been seen in other industries, effective implementation of AI can lead to significant long-term cost savings.⁶ Previously literature exploring AI implementation in health care has been scarce. However, in recent years the area has seen a rapid increase in investigation. Many studies have been able to provide good evidence of the efficacy of AI, but real-world implementation studies are still rare. There is an inherent need for care to be taken with AI implementation, and barriers continue to exist at every level from patient factors to government policymakers.⁷ At the policymaker level, economic implications of implementation play a large role in government adoption of new technology.

The current application of AI in ophthalmology is centered around large-scale screening programs for highly prevalent conditions including diabetic retinopathy and age-related macular degeneration (ARMD).⁸ The most common current implementation is the use in diabetic retinopathy

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referral pathways to stratify the need for further human assessment and treatment.⁷ Studies examining AI in these settings largely affirm the efficacy of this technology as comparable to that of human clinicians.⁹ Conclusive data on the economics of AI implementation in the above settings and other areas in ophthalmology are scarce and therefore further investigation is needed to determine whether AI is cost-effective. This systematic review aimed to evaluate previous economic analyses of the implementation of AI in clinical ophthalmology, with a particular focus on the Australian health care setting.

METHODS

This review was conducted in accordance with the preferred reporting items for systematic reviews and meta-analyses 2020 guidelines (Supplementary Digital Content Appendix 1, <http://links.lww.com/APJO/A175>). We conducted a comprehensive literature search through the PubMed/MEDLINE, EMBASE, and CENTRAL databases from inception to March 1, 2022. We searched references and clinical trials registries listed in relevant reports. We searched using the keywords and search terms “ophthalmology,” “artificial intelligence,” “AI,” “deep learning,” “economic evaluation,” “cost,” “cost-effectiveness,” “healthcare economics,” “cost-benefit analysis,” and “Australia.” Boolean operators OR/AND were employed to diversify the search terms, as detailed in Supplementary Digital Content Appendix 2, <http://links.lww.com/APJO/A176>.

To be included, an article must be primary research and the full-text must be available online. It must analyze a population of patients who have an ophthalmological diagnosis, or who are currently being evaluated for an ophthalmological diagnosis. It must use a health economic analyses (HEA) system to assess the cost-effectiveness of AI. A HEA was defined as a grading system that comparatively analyzes the costs of 2 health technologies, in this case AI versus manual clinician-based assessment. Included studies must also be published in English. Our initial search also filtered for studies conducted in an Australian setting; detailed in Supplementary Digital Content Appendix 2, <http://links.lww.com/APJO/A176>. Unfortunately, the current literature lacks any published studies on the topic specific to Australia, and as such the authors revised the search strategy to include studies from around the globe. The revised search strategy is shown in Supplementary Digital Content Appendix 3, <http://links.lww.com/APJO/A177>. Criteria for exclusion from our review were studies in which no HEA was used to assess cost-effectiveness. Risk of bias was then analyzed using the Johanna Briggs Institute Critical Appraisal Checklist for Economic Evaluations.¹⁰ Data were extracted regarding the AI model evaluated, HEA method used, and results of HEA evaluation. Eligibility determination, risk of bias evaluation, and data extraction were performed in duplicate (J.P. and A.L.) with instances of disagreement resolved by discussion with a third investigator (S.B.).

RESULTS

The search identified a total of 206 results: 188 results on PubMed/MEDLINE, 9 results from EMBASE, and 4 results from CENTRAL (Fig. 1). After screening of abstracts,

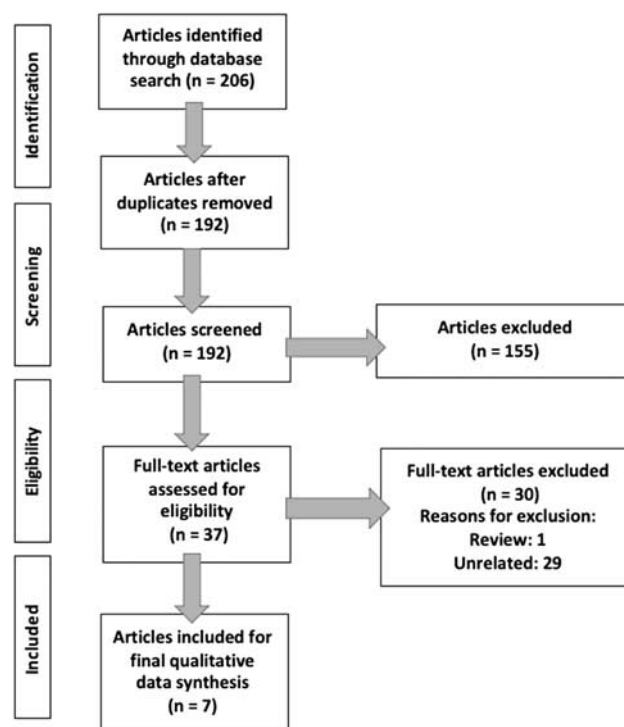


FIGURE 1. PRISMA flowchart illustrating flow of information during the search. PRISMA indicates preferred reporting items for systematic reviews and meta-analyses.

37 articles underwent full-text review. Ultimately, 7 articles were identified for inclusion (Table 1). All 7 included articles had a low risk of bias (Table 2).

Effectiveness of AI

AI was found to be at least equally as effective as clinician assessment across the 2 major ophthalmology applications analyzed: diabetic retinopathy and ARMD. Scotland et al¹¹ showed that the AI algorithm used in their study identified 86.9% of diabetic retinopathy cases it was presented with, comparable with the 87.7% identified by manual human grading. Fuller et al¹² echoed this finding in their 2022 study, which showed that the EyeArt 2.0 automated retinal imaging analysis system (ARIAS) was equally as effective as human grading for fundus photographs of diabetic retinopathy. Looking at ARMD, screening with AI was shown to reduce the incidence of eventual blindness due to ARMD in the analyzed population by 40.7%, when compared with patients who were not screened.¹³

Health Economic Analyses

Health economic analyses (HEA) can be conducted using a variety of methods. The articles included in our review utilized cost-utility analyses (1 study), cost-effectiveness analyses (CEAs) (4 studies), cost-saving analyses (1 study), and cost-minimization analyses (CMAs) (1 study) to assess the health economic implications of the ophthalmological implementation of AI.

Cost-Effectiveness Analyses

A CEA is a HEA that examines the costs and the effectiveness of a health care intervention.¹⁴ An intervention is compared with the gold standard or the normal, and the cost per unit of a favorable health outcome (eg, 1 y of life) is measured. This is termed the incremental cost-effectiveness

TABLE 1. Summary of Currently Available Literature on Health Economic Analyses of AI Implementation in Ophthalmology

References	Diagnosis and Sample Size	Model	HEA and Time Frame	Key Points	Strengths	Limitations
Fuller et al ¹²	DR 179	DT	CUA 5 y	ARIAS reduced costs by 23.3% vs manual grading. The calculated ICUR of \$258,721.81 comparing the current practice to ARIAS screening is well beyond the assessed willingness-to-pay threshold of \$100,000.	Results analyzed over a 5-year horizon	Only took into consideration screening for DR, not other ocular conditions. Converted visual acuity to QALY figures—QALY is an imperfect measure of a patient's quality of life. Model only included direct costs of screening and treatment from the payor's perspective and did not account for indirect costs to patients
Scotland et al ¹¹	DR 6722	DT	CEA 1 y	AI identifies 86.9% of cases and manual grading 87.7%, comparable effectiveness. The additional cost per referable case detected (manual vs AI) totaled £4088 and the additional cost per additional appropriate screening outcome (manual vs AI) was £1990. Cost per patient using AI must be £1.40 for AI to become more costly	Earliest economic evaluation of AI for DR screening for implementation in a national screening program	Only considers costs and consequences over a 1-year horizon. Only assess as disease versus no disease, disease severity not assessed. Analyzed machine learning AI—less accurate than the current deep learning AI systems
Tamura et al ¹³	ARMD Not specified	DT/ Markov Model	CEA 50 y (simulated)	When compared with the nonscreening group, the ICERs were ¥10,524,003/QALY, ¥10,437,363/QALY, and ¥10,491,265/QALY for screening with fundus photos interpreted by AI, screening with OCT interpreted by AI, and screening with OCT interpreted by a clinician, respectively	Intervention for ARMD assessed	Could not set the additional cost when implementing AI so assumed no additional in cost. Some potentially confounding factors not analyzed, for example, the proportion of patients dropping out, the effectiveness of supplements, and the rates and costs of subsequent examinations
Tufail et al ¹⁷	DR 20258	DT	CEA 1.5 y	Two strategies—initial grading undertaken by human grades with an ARIAS vs using ARIAS prior to manual grading ARIAS saved costs compared with manual grading both as a replacement for initial human grading and as a filter prior to primary human grading. Retmarker: \$6.04 per patient for strategy 1 and \$5.19 for strategy 2. EyeArt cost \$4.29 (strategy 1) and \$3.29 (strategy 2) per patient	Retmarker and EyeArt did not miss any vision threatening non-DR retinal conditions from the subset of images that went to the reading center for arbitration	The 2 strategies used were semiautomated not fully automated. Not immediately applicable as one ARIAS processes images using cloud technology and governance issues associated with the form of data storage needs to be addressed before implementation
Wang et al ²⁴	DR 88363	DT	CSA 1 y	The total time required for semiautomated DLA-assisted analysis of the follow-up data sets was 139 hours (0.09 min per image) and the total cost ~\$1544 (\$0.017 per image). This is an estimated 75.6% time saving and 90.1% cost saving compared with entirely human grading of the same dataset	Large sample size	Lingtau Cohort Study data sets used, which is representative only of the primary care setting within this region of China—limited generalizability. Some DR cases may have been missed as 5% of images were ungradable due to poor quality or poor positioning. Fundus images used in this study were 2-field, nonstereoscopic images that may detect less DR compared with gold standard 7-field images

Wolf et al ¹⁶	DR Not specified	DT	CEA 1 y	Current 20% adherence rate dictates that use of autonomous AI would result in a higher mean patient payment (\$8.52 for T1DM and \$10.85 for T2DM) than conventional human screening (\$7.91 for T1 and \$8.20 for T2). Adherence rate of 23% would make AI cheaper	Threshold of screening uptake required for cost-effectiveness (23%) is very close to the existing uptake rate of 20%	Only pediatrics assessed. Not fully automated as referrals were confirmed as DR by clinicians. Did not evaluate system or societal costs of technology implementation or personnel or office finances
Xie et al ²³	DR 39006	DT	CMA 1 y	A patient with diabetes would incur a 12-month total cost of \$77, \$62, and \$66 for the human assessment, semiautomated, and fully automated screening models, respectively	No statistical difference in terms of sensitivity across all the 3 models. The number of patients successfully referred and treated was assumed to not differ across the 3 models, justifying the cost-minimization HEA	Offset of costs determined by Singaporean wages—limited applicability worldwide. Exclusion of diabetic macular edema with mild referable DR could have underestimated cost-saving from fully automated model. Referrals based on referral guidelines in Singapore introduce referral bias

AI indicates artificial intelligence; ARIAS, automated retinal imaging analysis systems; ARMD, age-related macular degeneration; CEA, cost-effectiveness analysis; CMA, cost-minimization analysis; CSA, cost-saving analysis; CUA, cost-utility analysis; DLA, deep learning algorithm; DR, diabetic retinopathy; DT, decision tree; HEA, health economic analyses; ICER, incremental cost-effectiveness ratio; ICUR, incremental cost-utility ratio; OCT, optical coherence tomography; QALY, quality-adjusted life years; T1DM, type 1 diabetes mellitus; T2DM, type 2 diabetes mellitus.

ratio (ICER).¹⁵ A limitation of this HEA is that the outcomes compared must be disease specific. The utility of a CEA to compare interventions measuring different outcomes is limited unless a standardized outcome measure is used, such as quality-adjusted life years (QALYs).

Four of our included publications utilized a CEA as their HEA of choice. Wolf and colleagues in 2020 analyzed the cost-effectiveness of AI technology in diabetic retinopathy screening in a cohort of pediatrics with a mix of type 1 and type 2 diabetes mellitus. The researchers found that the average out of pocket cost to patients screened manually was \$7.91 for patients with type 1 diabetes and \$8.20 for those with type 2 diabetes. Conversely, type 1 diabetes patients screened with AI were out of pocket \$8.52, and type 2 diabetes patients \$10.85. These results assumed the average screening uptake rate in the American population of 20%. ICER analysis determined a screening uptake rate of 23% would be required to make AI screening more cost-effective than manual screening—a relatively small increase on the current rate.¹⁶

A 2017 study by Tufail and colleagues similarly analyzed the cost-effectiveness of 2 AI systems, Retmarker and EyeArt, compared with human grading of diabetic retinopathy in an adult population. Two screening strategies were analyzed in this study. Strategy 1 assessed AI as a precursor to human screening, and strategy 2 assessed AI as a replacement to human screening. The measured outcome was the percentage of patients missed by the AI systems with potentially sight-threatening diabetic retinopathy. The results showed that using an AI system with both strategies was more cost-effective than using human grading alone. Both the Retmarker and EyeArt systems were more cost-effective using strategy 1, saving \$15.36 more per appropriate outcome when compared with strategy 2. Threshold analysis using ICER was performed to determine the cost per patient at which AI would become more expensive than human grading alone. Using Retmarker, this point was identified as \$6.04 under strategy 1 and \$5.19 under strategy 2. Comparatively, the threshold was \$4.29 under strategy 1 and \$3.29 under strategy 2 using EyeArt.¹⁷ It should be noted that although these results are promising, they may not be immediately applicable on a large scale as one of the AI systems utilizes cloud-based storage for its data, which comes with associated governance and confidentiality issues.

Scotland and colleagues in 2007 published arguably the first study on the topic, analyzing the cost-effectiveness of implementing AI screening for diabetic retinopathy into the National Health Service (NHS) in Scotland. Outcomes were defined as recall for rescreening in 6 months, recall in 12 months, or referral to an ophthalmologist. The additional cost of each referable case of diabetic retinopathy detected using manual grading compared with AI was £4088, with the additional cost per any appropriate screening outcome of the 3 detected being £1990. The authors concluded that implementing the automated AI system would lead to savings of £201,600 annually. Interestingly, sensitivity analysis of the results showed that automated AI screening would still result in savings of £179,200 to the NHS annually if implementation costs were doubled, emphasizing the cost-effectiveness of this technology. Threshold analysis determined the point at which AI would become less cost-effective than manual grading to

TABLE 2. Johanna Briggs Institute Risk of Bias Analysis for Included Articles

JBIC—Critical Appraisal Checklist for Economic Evaluations	Xie et al²³	Tufail et al¹⁷	Wolf et al¹⁶	Scotland et al¹¹	Fuller et al¹²	Tamura et al¹³	Wang et al²⁴
Is there a well-defined question/objective?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Is there a comprehensive description of alternatives?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Are all important and relevant costs and outcomes for each alternative identified?	Yes	Yes	No	Yes	No	No	Unclear
Has clinical effectiveness been established?	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Are costs and outcomes measured accurately?	Unclear	Unclear	Unclear	Unclear	Unclear	Unclear	Yes
Are costs and outcomes valued credibly?	Yes	Yes	Yes	Yes	Yes	Yes	Unclear
Are costs and outcomes adjusted for differential timing?	Yes	Yes	No	No	Yes	No	No
Is there any incremental analysis of costs and consequences?	Yes	Yes	Yes	Yes	Yes	Yes	No
Were sensitivity analysis conducted to investigate uncertainty in estimates of costs or outcomes?	Yes	Yes	Yes	Yes	Yes	Yes	No
Do study results include all issues of concern to users?	Yes	Unclear	No	No	Unclear	No	No
Are the results generalizable to the setting of interest in the review?	No	Unclear	Unclear	Unclear	Unclear	No	Unclear
Overall rating	Low risk	Low risk	Moderate risk	Low risk	Low risk	Moderate risk	High risk

JBIC indicates Johanna Briggs Institute.

be £1.40 per patient screened.¹¹ This study is limited by the fact it only assessed costs over 1 year, leaving longer-term consequences uncertain. It also assessed diabetic retinopathy cases as disease-present or disease-absent and did not consider disease severity and the impact this may have on referable case numbers.

The last CEA that was included in our review was that of Tamura and colleagues, analyzing the cost-effectiveness of AI use in ARMD screening. The measured outcome in this analysis was the cost of screening per additional QALY gained. The ICER was compared between groups who received no screening, who were screened with fundus photography interpreted by AI, who were screening with optical coherence tomography (OCT) interpreted by AI, and who were screened with OCT interpreted by an ophthalmologist. The ICERs were ¥9,846,411/QALY, ¥10,524,003/QALY, ¥10,437,363/QALY, and ¥10,491,265/QALY, respectively. Naturally, no screening at all was the least resource-intensive and therefore the most cost-effective. Of the screening strategies, OCT interpreted by AI was the most cost-effective and fundus photography interpreted by AI the least cost-effective.¹³ There were significant limitations that must be acknowledged when interpreting the results of this study. The impact on the results of parameters including when disease states differ in each eye of a patient, the proportion of patients who do not follow-up with treatment, and the costs of subsequent periodic ophthalmologist examinations were not considered. Thus, cost projections from these variables require additional speculation which may render the results inaccurate.

The 4 CEA models included all suggest autonomous AI may be a more cost-effective screening strategy than traditional

manual grading. This is shown in both adult and pediatric population, and across diabetic retinopathy and ARMD screening programs. It should be noted that all the above studies were conducted in first-world, well-resourced countries and thus the applicability of these results to countries with less-advanced health care systems is questionable. First-world health care systems may also already have appropriate infrastructure and implementation know-how, while poorly developed nations may incur additional set up and training costs detracting from the overall cost-effectiveness of AI. The major limitation of these analyses was that patients were not analyzed over a lifetime, leaving total costs to patients unknown. This may be better evaluated using a cost-utility analysis (CUA).

Cost-Utility Analyses

CUA enables investigators to compare a wider range of interventions than CEA using the same outcome—cost per QALY gained. CUA is the superior HEA method in several situations. These include when many outcomes result from an intervention and a single generalizable measure is needed, when the intervention impacts morbidity over time as well as mortality, and finally when quality of life is a crucial metric of an intervention comparison.¹⁸ The major criticism of CUA is based on its use of QALY as an outcome measure. QALY is an imperfect measure has been criticized for its lack of applicability to the health care provider, and bias toward younger people as the elderly are deemed to have limited years left to impact the measurement. In the US, the Patient Protection and Affordable Care Act has forbidden QALY from being used as a threshold to determine the cost-effectiveness of new and emerging medical interventions.^{19,20}

One of the studies included in our review used a CUA to evaluate the economic implications of AI in ophthalmology. Fuller and colleagues examined the effectiveness of AI analysis of retinal imaging as part of a screening program for diabetic retinopathy across a 5-year span. The 2 AI systems currently approved for use in ophthalmology by the US Food and Drug Administration (FDA), IDx-DR and EyeArt, were compared with manual screening by clinicians to determine the most cost-effective method. The authors found that both AI systems reduced costs by 23.3% per patient when compared with manual grading. Threshold analysis determined that this effect would remain the case until the population screening uptake rate reached 71.5%.¹² The willingness-to-pay threshold is a metric designed to assess the maximum price a member of a population is willing to pay to benefit from an outcome.²¹ At a willingness-to-pay of \$50,000, the model favored AI in 66% of instances, and AI was most cost-effective in 59% of cases when the willingness-to-pay was set at \$100,000.¹² It is important to note that this study did not consider indirect costs to patients including transport costs or forced sick leave, likely underestimating the true cost to patients.

Cost-Minimization Analyses

Xie and colleagues conducted an analysis of the viability of AI for teleophthalmology-based diabetic retinopathy screening using a CMA model. The CMA model estimates the cost of an intervention compared with other interventions with comparable efficacy, considering all costs inherent to the delivery of the intervention.²² Similarly to other HEAs previously discussed, it aims to identify the least costly implementation strategy. The authors compared traditional human grading of diabetic retinopathy with semiautomated AI and fully automated AI systems for the screening of diabetic retinopathy in a Singaporean population. The outcome assessed was cost over a 12-month period. Patients incurred a 12-month total cost of SGD77, SGD62, and SGD66 using human grading, semiautomated AI, and fully automated AI systems, respectively. Compared with human grading, cost reductions of SGD15 (19.5%) for semiautomated AI and SGD11 (14.3%) for fully automated AI were seen. The higher cost seen with fully automated AI compared with semiautomated was postulated to be due to a higher rate of false-positives leading to more unnecessary specialist visits. These results were extrapolated to estimated savings of SGD489,000 to the Singaporean public health system annually—approximately 20% of the current yearly screening cost for diabetic retinopathy. By 2050, this figure is expected to rise to potential savings of SGD15 million. The costs included in this study were estimated using average Singaporean wages, limiting the applicability of the results globally.

Cost-Benefit Analyses

Cost-benefit analysis (CBA) is universally considered the most accurate and comprehensive HEA method. It values the impact of interventions on monetary units based on the traditional welfare economics theory.¹⁸ This allows ease of comparison between interventions and easy justification of implementation for policymakers. The use of CBA in the current literature has been limited due to the difficulty in

assigning monetary units to arbitrary imperfect outcome measures such as QALY.²³

In 2021, Wang and colleagues published a study analyzing the cost-effectiveness of the use of semiautomated AI as part of a diabetic retinopathy screening program in a Chinese population. This is one of the only currently available studies which has analyzed the health economic implications of AI in ophthalmology utilizing a CBA. The investigators measured outcomes using time and cost metrics. The total screening time required per case was 0.09 minutes per image using semiautomated AI, compared with the 0.16 minutes required for human screening. This is a time saving of 75.6%. From a cost perspective, the total cost per retinal image assessed by AI was \$0.017 versus \$0.032 per image graded by humans. This result represents a 90.1% cost saving.²⁴ A large sample size of 88,363 images was assessed, further validating the results. The minimal available CBA evidence on the topic indicates the use of AI in ophthalmology is an economically viable practice with efficacy comparable to human grading.

DISCUSSION

This systematic review is the first to characterize the global literature regarding AI implementation in ophthalmology viewed from an Australian perspective. Our study suggests that the clinical effectiveness of AI is equal to that of human clinicians, but more primary health economic analyses are needed to determine its cost-effectiveness in Australia. Our review identified 4 studies which utilized CEA and all showed AI to be more cost-effective than existing human screening strategies in the diabetic retinopathy context. One study examining ARMD also suggested superior cost-effectiveness and clinical effectiveness for AI over traditional human grading. These results show promise for AI as a cost-effective implementation in ophthalmology going forward. There exists no primary literature examining AI use in Australian ophthalmology, and this remains a potential barrier to widespread implementation.

Costs of AI

A variety of methods exist for analyzing costs in health care. One approach is that from the perspective of the health care provider. This includes the analysis of costs of hardware and software purchase, as well as maintenance costs.²⁵ These costs are balanced with time costs of clinicians to evaluate the viability of implementation. The gold standard, however, is cost-analysis from the perspective of the community.²⁵ Using this method, the impact of factors such as transport cost, follow-up appointments, and treatment cost are considered in addition to costs incurred by governments and health care providers.¹⁰ This evaluation gives a more complete picture of the total economic impact of implementation. Despite this importance, societal costs are often difficult to generalize and as such estimates are often assumed, or costs are analyzed from the provider perspective.

AI development is universally recognized as a potentially expensive endeavor. When used in health care, it has been categorized a medical device by the FDA.²⁶ In most screening and diagnostic situations in health care, the medical device constitutes the majority of costs incurred by the health care

provider. In the ophthalmology setting however, AI software use is often paired with use of equally expensive hardware, such as OCT scanners, slitlamps, and devices to facilitate high-definition fundus photography.²⁶ In many cases the cost of this hardware may exceed the cost of AI software installation. It is important to note that the same hardware is employed by human clinicians when grading diabetic retinopathy and ARMD patients, and as such this cost is eliminated in economic evaluations of AI in ophthalmology.²⁵ In addition, it is important to mention that total costs of AI in health care are poorly defined in the existing literature. Initial short-term costs of AI include purchase of AI software as well as integration into the existing health care computer system. Hardware purchase may also be required to facilitate AI use in some settings where this hardware was not previously available. Over the medium-term, data processing and transit to clinicians for further assessment and follow-up of pathology further add to total costs. The cost of software maintenance, updates, and troubleshooting needs to be considered over the long-term. Research and development costs of AI prior to installation may also be passed on to providers, although this is contentious as providers do not directly pay these costs outright.²⁶

In the US, 2 AI systems have been approved by the FDA for diabetic retinopathy screening in ophthalmology—the IDx-DR and EyeArt.²⁷ In 2020, Chen and colleagues analyzed the cost of the IDx-DR system and found the cost to be \$13,000 initially, plus an additional \$25 per patient screened. A negative revenue of \$1.18 would be seen with the application of the US Medicare system patient reimbursement, making IDx-DR a costly exercise.²⁸ The cost of the EyeArt system has not been widely analyzed, as it only gained approval in the US in late 2020. Further AI screening devices may be approved for use in ophthalmology in the near future and it remains to be seen whether pricing schemes will be comparable between systems. Australia's Therapeutic Goods Administration (TGA) is yet to approve any AI screening technology for use in clinical ophthalmology.²⁹

Until recently, there was a well-recognized shortage of publications exploring the economic viability of AI in the health care industry. In 2020, Wolff and colleagues concluded that there were very few studies exploring economic viability of AI available on scientific databases worldwide. In addition, of those few that were available, several employed severely flawed methodology and analysis techniques. It was recommended that more comprehensive economic analyses be conducted to adequately inform policymakers as to whether AI should be implemented in health care.¹⁰ Since the publication of Wolff's review, there has been an explosion of research in this area and many more published studies are now available for analysis. Notably, ophthalmology has been one of the leading areas of application of AI in the health care sector in recent years. Despite this increase in AI research, there remains limited published health economic analyses in this area outside of a few specific applications, namely diabetic retinopathy and ARMD screening.

An Australian Perspective

Of the studies included in our review, 3 reported results in the context of the American health care system. Results can be

extrapolated to determine an estimated economic impact of AI implementation in ophthalmology in Australia. The most applicable of these are provided by Wolf and colleagues. From these results, it can be estimated that the average out of pocket costs for children screened for diabetic retinopathy with AI would be A\$6.63, compared with A\$5.51 for those children screened manually. With diabetic retinopathy screening uptake estimated at 50% to 77% in Australia, AI would likely become more cost-effective than manual grading.³⁰ Given the original health care expenditure figures originated from a 2018 report of the Organisation for Economic Co-operation and Development (OECD), all figures assumed an average exchange rate of \$1 = A\$1.34 in 2018.³³

Increased specificity of current medical screening methods has resulted in so-called “medical overuse.” This includes increased unnecessary testing, diagnosis, and treatment; all of which increase economic and resource burdens on both patient and provider. For example, 50% of preoperative testing for cataract surgery is deemed unnecessary due to associated surgical and anesthetic risks for the individual.³⁴ AI can also be used on a larger scale in the management of health systems by monitoring health expenditure, recovery costs, and treatment response, as well as identifying areas of both overuse and underuse of the medical system.

In order to decrease overtesting and subsequent overdiagnosis, AI can be used to increase the personalization of health care for the individual. Analyzing each case on its merits, rather than following predetermined referral guidelines often in use in Australian health care services, has the potential to reduce unnecessary referrals to tertiary services. The result is improved ability to analyze each case quickly and efficiently on a large scale in sometimes under-resourced settings and maximizing subsequent savings.

Rural Aspect

AI has been used to create novel ocular imaging modalities, which include slitlamp adapters and retinal cameras attached to smartphones. Such systems have the potential to facilitate better outreach of cataract screening programs, specifically in rural areas. This is particularly pertinent in a country such as Australia, with a large percentage of our land mass classified as rural. Such systems could be operated by a trained technician or nurse instead of a specialist, reducing associated manpower costs while making it more readily in a rural environment where specialists are scarce.

Current available AI-based screening programs, for example, retinal photography assessing diabetic retinopathy, may simultaneously serve as an opportunistic screening tool for cataract detection with minimal additional cost to patient or provider.³⁵

A significant consideration in the Australian setting is the prevalence of rural populations. Rural patients, who are often referred to ophthalmology services based on initial testing, must often travel a significant distance to tertiary hospitals. The use of AI in combination with teleophthalmology has potential to reduce the burden of false-positive referrals and those triaged to tertiary care. This thereby reduces costs associated with unnecessary travel, relieving burdens both economically and on an already resource-strained health care system.³⁶

Limitations

It is important to note the limitations of extrapolating global data in the absence of primary Australian data. The 2 FDA-approved AI software systems are both produced in the US, meaning importation costs would heavily impact eventual cost per patient of employing this technology in Australia. Government levies applied to imported goods may also further increase costs. Although the health care system of Australia and the US are comparable in some areas, key differences exist that limit the utility of large generalizations. Health rebate rates differ between the countries, and health care systems such as Medicare constantly revise the amount they subsidize. On a community level, indirect costs incurred to patients may vary significantly on an international scale. The costs of specialist consultation, transport, forced sick leave/unpaid leave all differ between locations, and hence it is difficult to speculate on the impact of these costs in the Australian landscape. Health care referral guidelines may also differ. This difference is particularly relevant to screening applications of AI. Different thresholds for referability of a diabetic retinopathy case, for example, may impact the total number of referable cases and ultimately mean more patients are subjected to higher out of pocket costs of the long-term. Another limitation of this review is the potential impact of publication bias. With the worldwide rise in AI use, the articles included in our review suggest a positive potential cost-effectiveness of AI. Limited publications exist which evaluate the negative economic effectiveness of using AI, highlighting potential publication bias.

For 6 out of 7 studies included in the review, it was difficult to determine whether the costs and outcomes were measured accurately. This was primarily due to the poorly detailed cost measurement strategies outlined within the methods sections of these studies. This is acknowledged as a limitation, as the indirect impact of these poorly defined strategies on this systematic review is difficult to quantify. It must also be noted that only studies published in English were included, potentially limiting relevant foreign language studies from inclusion.

Future Research

There are multiple areas for further research in the health economic analyses of ophthalmology AI applications. In particular, further CBA are required in diverse health care settings, such as Australia. Markov models are employed in health care economics to analyze stochastic processes, for example, processes that randomly evolve over time, such as disease processes. This allows economic analysis of an intervention considering the impact of evolving disease severity over time.³¹ In our review, only 1 study analyzing diabetic retinopathy used a Markov Model in addition to a decision tree in their economic analysis. Future studies should consider the evolving disease state over time and the resulting health economic implications when conducting analysis.

Other areas of ophthalmology also warrant economic analysis of AI use, including glaucoma, ARMD, and cataract. These areas constitute some of the largest contributors to blindness across the world and have as much potential for AI use as diabetic retinopathy. There exist early prototypes of AI

screening software for both cataract and glaucoma using color fundus photography, but it is important to note that none of these have been subject to HEA and remain in the initial stages of clinical effectiveness determination.^{32,33} HEA of these settings may lead to AI implementation across a wider cross-section of ophthalmology.

AI is a versatile technology with significant health economic implications. Diabetic retinopathy is by far the most well-researched application of AI in ophthalmology, albeit with still a small number of publications on the topic. A common limitation among the studies included in our review was relatively short follow-up timeframes. Future studies may benefit from lifetime economic analysis of AI in the diabetic retinopathy context, providing clearer evidence as to AI's economic viability in this setting.

Recently, the American health care system has begun to introduce billable codes, for example, code CPT 92229, for reimbursement for autonomous AI services provided to ophthalmology patients.³⁷ This raises the question of appropriate government reimbursement for ever-evolving technologies such as AI. Focusing on the US, the American Health Care System is traditionally funded through a “fee-for-service” model. The nature of this model makes it inherently difficult to measure the quality of service provided, as the issue becomes how to fund a technology that improves the baseline quality of the service provided. This compounds the already complicated undertaking of economic analyses in this area.

CONCLUSIONS

Our review emphasizes the need for primary economic evaluation of AI in ophthalmology in an Australian setting. While ongoing overseas research illustrates the potential economic viability of AI in ophthalmology, such positive results cannot be immediately applied to an Australian setting. The difference in currency exchange rates, importation of technology, and differences in health care system reimbursement all add to the limited external generalizability of foreign health economic analyses. To enable accurate modeling of costs per patient, it is essential for future studies to take such factors into consideration.

REFERENCES

1. World Health Organization. Global Health Expenditure Database. World Health Organization. 2022. Available at: <https://apps.who.int/nha/database>. Accessed March 31, 2022.
2. Organisation for Economic Co-operation and Development. OECD Health Statistics 2021. Organisation for Economic Co-operation and Development. 2021. Available at: <https://www.oecd.org/els/health-systems/health-data.htm>. Accessed March 31, 2022.
3. Parliament of Australia. Health. Parliament of Australia. 2019. Available at: https://www.aph.gov.au/About_Parliament/Parliamentary_Departments/Parliamentary_Library/pubs/rp/BudgetReview201920/Health. Accessed March 31, 2022.
4. Organisation for Economic Co-operation and Development. Health Spending Indicator. Organisation for Economic Co-operation and Development. 2021. Available at: <https://data.oecd.org/healthres/health-spending.htm>. Accessed March 31, 2022.

5. Xiang Y, Zhao L, Liu Z, et al. Implementation of artificial intelligence in medicine: Status analysis and development suggestions. *Artif Intell Med*. 2020;102:101–780.
6. Singh RP, Hom GL, Abramoff MD, et al. Current challenges and barriers to real-world artificial intelligence adoption for the healthcare system, provider, and the patient. *Transl Vis Sci Technol*. 2020;9:1–6.
7. Du XL, Li WB, Hu BJ. Application of artificial intelligence in ophthalmology. *Int J Ophthalmol*. 2018;11:1555–1561.
8. Grzybowski A, Brona P, Lim G, et al. Artificial intelligence for diabetic retinopathy screening: a review. *Eye*. 2020;34:451–460.
9. Wolff J, Pauling J, Keck A, et al. The economic impact of artificial intelligence in health care: systematic review. *J Med Internet Res*. 2020;22:16866.
10. Johanna Briggs Institute. Checklist for economic evaluations. *JBIGlobal*. 2020. Available online: <https://jbi.global/critical-appraisal-tools>. Accessed April 3, 2022.
11. Scotland GS, McNamee P, Philip S, et al. Cost-effectiveness of implementing automated grading within the national screening programme for diabetic retinopathy in Scotland. *Br J Ophthalmol*. 2007;91:1518–1523.
12. Fuller SD, Hu J, Liu JC, et al. Five-year cost-effectiveness modelling of primary care-based, nonmydriatic automated retinal image analysis screening among low-income patients with diabetes. *J Diabetes Sci Technol*. 2022;16:415–427.
13. Tamura H, Akune Y, Hiratsuka Y, et al. Real-world effectiveness of screening programs for age-related macular degeneration: amended Japanese specific health checkups and augmented screening programs with OCT or AI. *Jpn J Ophthalmol*. 2022;66:19–32.
14. Office of the Associate Director for Policies & Strategy. Cost-Effectiveness Analysis. Centres for Disease Control and Prevention. 2021. Available at: <https://www.cdc.gov/policy/polaris/economics/cost-effectiveness/index.html>. Accessed April 7, 2022.
15. York Health Economics Consortium. Incremental Cost Effectiveness Ratio (ICER). York Health Economics Consortium. 2016. Available at: <https://yhec.co.uk/glossary/incremental-cost-effectiveness-ratio-icer/>. Accessed April 7, 2022.
16. Wolf RM, Channa R, Abramoff MD, et al. Cost-effectiveness of autonomous point-of-care diabetic retinopathy screening for paediatric patients with diabetes. *JAMA Ophthalmol*. 2020;138:1063–1069.
17. Tufail A, Rudisill C, Egan C, et al. Automated diabetic retinopathy image assessment software: diagnostic accuracy and cost-effectiveness compared with human graders. *Ophthalmology*. 2017;124:343–351.
18. Rudmik L, Drummond M. Health economic evaluation: important principles and methodology. *Laryngoscope*. 2013;123:1341–1347.
19. Kind P, Lafata JE, Matuszewski K, et al. The use of QALYs in clinical and patient decision-making: issues and prospects. *Value in Health*. 2009;12:27–30.
20. Neumann PJ, Weinstein MC. Legislating against use of cost-effectiveness information. *N Engl J Med*. 2010;363:1495–1497.
21. Varian HR. Microeconomic analysis. In: Varian HR, ed. *Microeconomic Analysis Vol 3*, 1st ed. W.W. Norton: New York 1992.
22. Rai M, Goyal R. Cost-minimization analysis—pharmacoeconomics in healthcare. In: Vohora D, Singh G, eds. *Pharmaceutical Medicine and Translational Clinical Research Vol 1*, 1st ed. Elsevier Science Publishing Co Inc. Amsterdam; 2018.
23. Xie Y, Gunasekaran DV, Balaskas K, et al. Health economic and safety considerations for artificial intelligence applications in diabetic retinopathy screening. *Transl Vis Sci Technol*. 2020;9:22.
24. Wang Y, Shi D, Tan Z, et al. Screening referable diabetic retinopathy using a semi-automated deep learning algorithm assisted approach. *Front Med (Lausanne)*. 2021;8:740–987.
25. He J, Baxter SL, Xu J, et al. The practical implementation of artificial intelligence technologies in medicine. *Nat Med*. 2019;25:30–36.
26. Westerheide F The artificial intelligence industry and global challenges. *Forbes*. 2019. Available at: <https://www.forbes.com/sites/cognitiveworld/2019/11/27/the-artificial-intelligence-industry-and-global-challenges/?sh=565e73313deb>. Accessed April 7, 2022.
27. US Food & Drug Administration. Medical Device Databases. US Food & Drug Administration. 2022. Available at: <https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/medical-device-databases>. Accessed April 7, 2022.
28. Chen EM, Chen D, Chilakamarri P, et al. Economic challenges of artificial intelligence adoption for diabetic retinopathy. *Ophthalmology*. 2020;128:475–477.
29. Therapeutic Goods Administration. The Australian Register of Therapeutic Goods. Therapeutic Goods Administration. 2021. Available at: <https://www.tga.gov.au/searching-australian-register-therapeutic-goods-artg>. Accessed April 7, 2022.
30. Foreman J, Keel S, Xie J, et al. Adherence to diabetic eye examination guidelines in Australia: the National Eye Health Survey. *Med J Aust*. 2017;206:402–406.
31. Exchange Rates.org.uk. US Dollar to Australian Dollar Spot Exchange Rates for 2018. Exchange Rates.org.uk. 2022. Available at: <https://www.exchangerates.org.uk/USD-AUD-spot-exchange-rates-history-2018.html>. Accessed April 8, 2022.
32. Briggs A, Sculpher M. An introduction to Markov modelling for economic evaluation. *Pharmacoeconomics*. 1998;13:397–409.
33. Li Z, He Y, Keel S, et al. Efficacy of a deep learning system for detecting glaucomatous optic neuropathy based on colour fundus photographs. *Ophthalmology*. 2018;125:1199–1206.
34. Chen CL, Lin GA, Bardach NS, et al. Preoperative medical testing in Medicare patients undergoing cataract surgery. *N Engl J Med*. 2015;372:1530–1538.
35. Goh JH, Lim ZW, Fang X, et al. Artificial intelligence for cataract detection and management. *Asia-Pac J Ophthalmol*. 2020;9:88–95.
36. Gunasekaran DV, Wong TY. Artificial intelligence in ophthalmology in 2020: a technology on the cusp for translation and implementation. *Asia-Pac J Ophthalmol*. 2020;9:61–66.
37. Medicare Coverage Database. Billing and coding: remote imaging of the retina to screen for retinal diseases. CMS.gov. Available at: <https://www.cms.gov/medicare-coverage-database/view/article.aspx?articleid=58914>. Accessed June 22, 2022.